(a) C Program to Find Largest and Smallest Element in an Array

Algorithm

1. Start
2. Read size of array.
3. Read array elements.
4. Assume first element is largest and smallest.
5. Compare every element.
6. Update largest if current element is bigger.
7. Update smallest if current element is smaller.
8. Print largest and smallest.
9. Stop.

Program:

```
#include <stdio.h>


int main()

{

   int a[100], n, i;

   int largest, smallest;


   printf("Enter number of elements: ");

   scanf("%d", &n);


   printf("Enter array elements:\n");


   for(i=0;i<n;i++)

   {

      scanf("%d",&a[i]);

   }
```

```
    largest = a[0];
    smallest = a[0];

    for(i=1;i<n;i++)
    {
        if(a[i] > largest)
        {
            largest = a[i];
        }

        if(a[i] < smallest)
        {
            smallest = a[i];
        }
    }

    printf("Largest = %d\n", largest);
    printf("Smallest = %d\n", smallest);

    return 0;
}
```

Sample Output:

Enter number of elements: 5

Enter array elements:
10

25
8
40
15

Largest = 40
Smallest = 8

(b) C Program to Count Vowels and Consonants

Algorithm

1. Start
2. Read a string.
3. Initialize vowel = 0 and consonant = 0.
4. Read each character.
5. If character is vowel → increment vowel.
6. Else if alphabet → increment consonant.
7. Print counts.
8. Stop.

Program

```c
#include <stdio.h>

int main()
{
   char str[100];
   int i;
   int vowels = 0, consonants = 0;

   printf("Enter a string: ");
   fgets(str, sizeof(str), stdin);

   for(i=0; str[i]!='\0'; i++)
   {
      if(str[i]=='a'||str[i]=='e'||str[i]=='i'||str[i]=='o'||str[i]=='u'||
        str[i]=='A'||str[i]=='E'||str[i]=='I'||str[i]=='O'||str[i]=='U')
```

```
        {
          vowels++;
        }
        else if((str[i]>='a' && str[i]<='z') ||
              (str[i]>='A' && str[i]<='Z'))
        {
          consonants++;
        }
    }

    printf("Vowels = %d\n", vowels);
    printf("Consonants = %d\n", consonants);

    return 0;
}
```

Sample Output

Enter a string:
Computer

Vowels = 3
Consonants = 5

Step-by-Step Execution

Input

Computer

Character	Type	Vowels	Consonants
C	Consonant	0	1
o	Vowel	1	1
m	Consonant	1	2
p	Consonant	1	3
u	Vowel	2	3

Character	Type	Vowels	Consonants
t	Consonant	2	4
e	Vowel	3	4
r	Consonant	3	5

Final Output

Vowels = 3
Consonants = 5